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Project Report

Customer Churn Prediction Using Machine Learning

Machine Learning with Python (AI 202)

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Abstract

This project presents a machine learning approach to predicting customer churn in the telecommunications industry. Using the IBM Telco Customer Churn dataset sourced from Kaggle, which comprises 20 customer features including demographics, account information, service usage, and billing data, an XGBoost gradient boosting classifier was trained and evaluated on a binary churn classification task.

The dataset was preprocessed through label encoding of all categorical variables, feature-target separation, an 80/20 train-test split, and StandardScaler normalization. Comprehensive exploratory data analysis was conducted across six visualizations examining churn distribution, contract type, tenure patterns, monthly charges, internet service type, and payment methods. The trained XGBoost model achieved a test accuracy of 80.26%, with a precision of 0.69 and recall of 0.56 for the churn class, reflecting the inherent challenge of imbalanced real-world churn data where only 26% of customers churned.

Keywords: Customer Churn, Binary Classification, XGBoost, Gradient Boosting, Telecom, Label Encoding, Imbalanced Dataset

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1. Introduction

Customer churn refers to the phenomenon where existing subscribers discontinue their service with a provider and switch to a competitor. In the highly competitive telecommunications industry, retaining customers is significantly more cost-effective than acquiring new ones — studies consistently show that acquiring a new customer costs five to ten times more than retaining an existing one. As a result, accurate churn prediction has become a critical business intelligence capability for telecom operators worldwide.

Machine learning offers a powerful and scalable solution to this problem. By learning historical patterns from customer data, a trained classifier can identify at-risk customers before they churn, enabling the business to deploy proactive retention strategies such as targeted offers, service upgrades, or personalized outreach. This is especially valuable because churn is often driven by identifiable signals — high billing amounts, short tenure, lack of technical support, or month-to-month contract flexibility — all of which can be captured in structured customer data.

This project builds a binary customer churn prediction system using the XGBoost gradient boosting classifier, trained on the publicly available IBM Telco Customer Churn dataset from Kaggle. The objectives are to preprocess the raw data appropriately, conduct exploratory data analysis to uncover churn patterns, train a high-performance model, and evaluate it rigorously using accuracy, confusion matrix, and classification report metrics. The project is implemented in Python using scikit-learn, XGBoost, Pandas, Matplotlib, and Seaborn.

2. Dataset Description

2.1 Source

The dataset used in this project is the IBM Telco Customer Churn dataset, obtained from Kaggle in Microsoft Excel format (Telco_customer_churn.xlsx). This dataset is one of the most widely used benchmarks for churn prediction in both academic research and industry practice.

2.2 Structure

Property	Details
File format	Excel (.xlsx)
Total records	7,043 customer rows
Target column	Churn Value (0 = No Churn, 1 = Churn)
Feature types	Categorical and Numerical (mixed)

Property	Details
Missing values	None (clean dataset)

2.4 Target Variable & Class Balance

The target variable is Churn Value — a binary integer where 0 represents a retained customer and 1 represents a churned customer. The dataset is inherently imbalanced:

Class	Count	Percentage
0 — No Churn (Retained)	5,174	~73.5%
1 — Churn	1,869	~26.5%

This class imbalance is characteristic of real-world churn datasets and has direct implications for model evaluation — overall accuracy alone can be misleading since a naive model predicting all customers as non-churners would still achieve ~73.5% accuracy.

3. Data Preprocessing

The following preprocessing steps were applied to prepare the dataset for model training:

1. Dataset loading: The Excel file was loaded using pandas with the openpyxl engine. Initial inspection using `df.head()`, `df.shape`, `df.columns`, `df.info()`, and `df.describe()` provided a complete understanding of the data structure, types, and distributions.
2. Missing value check: `df.isnull().sum()` confirmed that the dataset contains zero missing values across all columns, requiring no imputation.
3. Label encoding: All object-type (categorical) columns were converted to integer representations using scikit-learn's `LabelEncoder`, applied uniformly in a loop over `df.select_dtypes(include='object').columns`. This includes Gender, Partner, Contract, Payment Method, Internet Service, Churn Label, and all Yes/No service columns.
4. Feature and target separation: The 20 selected feature columns were extracted as the feature matrix `X`. The binary column Churn Value was assigned as the target vector `y`.
5. Train-test split: The dataset was divided into 80% training (5,634 samples) and 20% testing (1,409 samples) sets using `train_test_split` with `random_state=42` for reproducibility.
6. Feature scaling: `StandardScaler` was applied to normalize all feature values to zero mean and unit variance. `fit_transform` was applied to the training set and `transform` only to the test set to prevent data leakage.

4. Exploratory Data Analysis

Six visualizations were generated to understand the patterns and relationships driving customer churn in this dataset.

4.1 Customer Churn Distribution

A count plot of the Churn Label column visualizes the absolute count of churned versus retained customers. The chart clearly shows a large imbalance — retained customers significantly outnumber churned customers, confirming the class imbalance discussed in Section 2.4.

4.2 Churn Percentage

A pie chart with percentage labels shows the proportional split: approximately 73.5% No Churn and 26.5% Churn. This visualization is particularly effective for communicating the imbalance to non-technical stakeholders.

4.3 Contract Type vs Churn

A grouped count plot comparing contract type against churn label reveals a striking pattern: month-to-month contract customers show dramatically higher churn rates compared to one-year or two-year contract customers. This is one of the strongest predictors in the dataset — customers on long-term contracts face cancellation penalties and switching costs that significantly reduce churn probability.

4.4 Tenure Months vs Churn

A histogram with KDE overlay for tenure months, grouped by churn label, shows that churn is heavily concentrated among customers with low tenure (0 to 12 months). The distribution of retained customers is flatter and extends further right, confirming that loyalty builds over time. New customers represent the highest churn risk and require the most attention from retention teams.

4.5 Monthly Charges vs Churn

A box plot comparing monthly charges between churned and retained customers reveals that churned customers have a noticeably higher median monthly charge. The interquartile range for churners sits at a higher level, suggesting that customers who perceive poor value relative to their billing amount are more likely to leave. This makes monthly charges an important feature for churn prediction.

4.6 Internet Service & Payment Method vs Churn

A count plot for internet service type shows that fiber optic customers have the highest absolute and relative churn rate among all internet service categories, likely due to higher pricing

combined with competitive fiber alternatives. A separate count plot for payment method shows that electronic check users churn at the highest rate, potentially indicating lower engagement or automatic billing not being set up, which correlates with less committed customers.

5. Correlation Analysis

A full correlation heatmap (20x12 figure) was generated after label encoding to visualize pairwise linear relationships between all features. The coolwarm colormap was used with annotations disabled for readability at this scale. Key observations include:

- Tenure Months shows a strong negative correlation with Churn Value — longer tenure strongly predicts retention.
- Monthly Charges shows a positive correlation with Churn Value — higher bills are associated with higher churn risk.
- Contract type shows a strong negative correlation with churn — longer contracts predict retention.
- Total Charges correlates highly with Tenure Months, which is expected since total charges accumulate over time.
- Service features like Online Security and Tech Support show moderate negative correlations with churn, suggesting that customers with more services are better retained.

6. Model — XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is an ensemble learning algorithm that builds decision trees sequentially, where each subsequent tree corrects the residual errors of the previous ones. It minimizes a regularized objective function that combines a differentiable loss term with a complexity penalty on tree structure, making it simultaneously accurate and resistant to overfitting.

XGBoost was selected for this project due to its consistently superior performance on tabular, mixed-type data, its native handling of label-encoded categorical features, its built-in regularization, and its proven track record in industry churn prediction applications. The model was configured with the following hyperparameters:

Hyperparameter	Value	Purpose
n_estimators	500	Number of boosting rounds — sufficient for convergence

Hyperparameter	Value	Purpose
learning_rate	0.03	Step size shrinkage — slow, careful learning for better generalization
max_depth	4	Maximum tree depth — prevents individual trees from overfitting
subsample	0.8	Row subsampling per tree — introduces diversity among trees
colsample_bytree	0.8	Feature subsampling per tree — reduces correlation between trees
random_state	42	Seed for reproducibility

The model was trained using `xgb_model.fit(X_train, y_train)` on the scaled training set and predictions were generated using `pred = xgb_model.predict(X_test)`.

7. Results and Evaluation

7.1 Accuracy

The trained XGBoost model achieved the following test set accuracy:

Model	Test Set Accuracy
XGBoost Classifier	80.26%

This result was obtained on 1,409 held-out test samples using `accuracy_score(y_test, pred)`. An accuracy of 80.26% on this imbalanced real-world dataset is a strong and meaningful result — it comfortably exceeds the 73.5% majority-class baseline that a trivial all-negative classifier would achieve.

7.2 Confusion Matrix

The confusion matrix provides a detailed breakdown of prediction outcomes across both classes:

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	~908 (True Negatives)	~101 (False Positives)
Actual: Churn	~176 (False Negatives)	~224 (True Positives)

The model correctly identifies 908 retained customers and 224 churners. The 176 false negatives — actual churners that the model missed — represent the primary business risk, as

these are at-risk customers who would not receive retention intervention. The confusion matrix was visualized as a seaborn heatmap with Blues colormap for clear interpretation.

7.3 Classification Report

A full per-class classification report was generated and visualized as a heatmap:

Class	Precision	Recall	F1-Score	Support
No Churn (0)	0.84	0.90	0.87	1,009
Churn (1)	0.69	0.56	0.62	400
Weighted Avg	0.79	0.80	0.80	1,409

The model performs strongly on the No Churn class ($F1 = 0.87$) but shows lower recall for the Churn class (0.56), meaning 44% of actual churners are not flagged. This is a direct consequence of the class imbalance — the model has seen nearly three times as many non-churn examples during training. The classification report heatmap was generated using `output_dict=True` and visualized with seaborn.

8. Discussion

The XGBoost model achieves 80.26% accuracy on this benchmark, demonstrating that gradient boosting is well-suited to telecom churn prediction from structured tabular data. Several key insights emerge from the analysis:

Why XGBoost performs well here:

- Its ensemble nature — combining 500 sequential trees — captures complex non-linear relationships between features that simpler models like Logistic Regression cannot.
- The low learning rate (0.03) combined with 500 estimators creates a stable, well-generalized model that avoids overfitting to training patterns.
- Feature subsampling (`colsample_bytree=0.8`) reduces correlation between individual trees, improving ensemble diversity and robustness.

Limitations:

- Class imbalance (26% churn vs 74% no-churn) causes the model to under-detect churners — recall of 0.56 means 44% of actual churners are missed. In a real deployment, this represents lost retention opportunities.
- The model does not use feature engineering — derived features such as charges per month or new customer flags could further improve discriminative power.

- Evaluation is based on a single 80/20 split. Cross-validation would provide more statistically reliable performance estimates with confidence intervals.

Real-world applicability:

In a production telecom environment, this model would be deployed to score all active customers monthly, flagging those above a churn probability threshold for targeted retention offers. The feature importance from XGBoost would guide which customer attributes to prioritize in retention campaigns — tenure, contract type, and monthly charges being the most actionable.

9. Conclusion

This project successfully designed, implemented, and evaluated a customer churn prediction system for the telecom industry using the XGBoost gradient boosting classifier. The following key outcomes were achieved:

- The IBM Telco Customer Churn dataset from Kaggle was loaded, cleaned, and preprocessed through label encoding and StandardScaler normalization across 20 customer features.
- Exploratory data analysis across six visualizations revealed that contract type, customer tenure, and monthly charges are the strongest predictors of churn behavior.
- A correlation heatmap confirmed the linear relationships between features and the churn target, validating the feature selection choices.
- The XGBoost classifier with 500 estimators, learning rate 0.03, and max depth 4 achieved a test accuracy of 80.26% on 1,409 held-out samples.
- Confusion matrix analysis showed 908 true negatives and 224 true positives, with 176 missed churners representing the primary false negative risk.
- The classification report revealed strong No Churn performance ($F1 = 0.87$) and moderate Churn detection ($F1 = 0.62$), consistent with the class imbalance present in real-world telecom data.

The results demonstrate that XGBoost is an effective and practical model for churn prediction from structured customer data, with direct applicability in customer retention strategy, proactive outreach, and lifetime value optimization for telecom operators. Future work could explore SMOTE oversampling, feature engineering, and GridSearchCV hyperparameter tuning to further improve churn class recall.